

# Taihe Interactive: Optimizing gaming experiences for gamers around the world with a highly connected global network

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ABOUT TAIHE



## About Taihe

Founded in 2014, Taihe Interactive is a game developer and distributor headquartered in Beijing specializing in action role-playing mobile games. Currently operating three popular games, Taihe Interactive has always been enhancing the gameplay, graphics quality, and aesthetics of its games for users around the world. Its flagship game Dynasty Legends 2 has reached more than one million downloads on Google Play with a 4.4-star rating.

Industries: Gaming

With the speedy ( network, the high performance virtu machines, as wel security and trans

capabilities of Google Cloud, Taihe Internet improves the gaming experiences that realizing better cooperation with more appealing features.

Location: China

(<https://cloud.google.com/load-balancing/>),

Google Cloud (<https://cloud.google.com/>)

(<https://cloud.google.com/storage/>), <https://cloud.google.com/compute/>

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

(<https://cloud.google.com/translate/>). Virtual

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Cloud Translation API

(<https://cloud.google.com/translate/>)

Virtual Private Cloud

(<https://cloud.google.com/vpc/>)

## Google Cloud results

- Improves connection stability and reduces network latency by more than 2X with Virtual Private Cloud
- Saves 30% of operational costs with custom VMs on

User cost down by 20% with better gaming experiences

**Compute**

**Engine**

- **Fully prevents DDoS attacks with Cloud Load Balancing**
- **Facilitates interactions among gamers worldwide and helps users understand game content better with Translation AI**

Action role-playing games have been one of the most popular genres of video games over the past two decades. As technology improves the performance of mobile devices, gamers also enjoy more exciting battle experiences on their smartphones or tablets. However, powerful IT infrastructures are required to support glitchless real-time combats. That's why mobile game developers are constantly looking for ways to optimize the stability and speed of their mobile gaming services.

## Taihe Interactive

(<https://www.taiyouxi.cn/index/about/index.html>) is a leading developer in the action role-playing mobile game category. Founded in 2014, the Beijing-headquartered game developer and distributor first gained its international fame through the release of

## Dynasty Legends

(<https://play.google.com/store/apps/details?id=com.taiyouxi.jwshmt&hl=en&gl=US>)

, a role-playing battle game in the backdrop of the Three Kingdoms period in ancient China with exciting gameplay and battling experiences. The sequel Dynasty Legends 2

(<https://play.google.com/store/apps/details?id=com.newtypegames.dl2hmt&hl=en&gl=US>)

came out in 2022 and has reached more than one million downloads on Google Play with a 4.4-star rating. Currently operating three popular mobile games, Taihe Interactive has millions of users around the world.

"Mobile gaming is a very competitive industry, with many games disappearing from the market after a few months. To build a strong brand, we're focusing on creating games that can last for a long time. That's why we have always been improving the gameplay, graphics and overall quality of our games," explains Dongping Zhang, Head of DevOps at Taihe Interactive.

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During our  
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virtual machine  
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and security,  
meeting all our  
requirements."

*—Dongping Zhang, Head  
of DevOps, Taihe  
Interactive*

To provide quality action gaming experiences to users around the world, having a stable and well-connected network is essential. However, when Taihe Interactive launched the global version of Dynasty Legends in 2017, users of the game frequently encountered long latency and network disruptions. This is because the game was deployed in five data centers worldwide that were not connected to each other. Users could only access a preassigned game server, instead of the server that's closest to them.

Besides network instability, Taihe Interactive's game servers also experienced an outage of 15 to 30 minutes every month, because the virtual machines (VMs) needed to be manually relaunched to cope with technical errors.

Having experienced these service stability and security issues, Taihe Interactive decided to launch Dynasty Legends 2 in another cloud environment to offer gamers a better gaming experience. In early 2022, the game developer landed on Google Cloud (<https://cloud.google.com/>), because it provides a speedy global network, high-performance VMs, and built-in security mechanisms against DDoS attacks.

"We learned from our previous experience that to improve our gaming services, we needed a more powerful cloud infrastructure," notes Zhang. "During our evaluation process, Google Cloud stood out in connectivity, virtual machine performance and security, meeting all our requirements."

## Enhancing user retention rate with a highly connected global network

With the technical support of the Google Cloud team, Taihe Interactive spent three months adapting its game to the architecture of the Google Cloud infrastructure and started running Dynasty Legends 2 on Google Cloud in March 2022. Now, it uses Virtual Private Cloud (<https://cloud.google.com/vpc>) to support global connectivity. Since all its game servers around the world on Google Cloud are connected to each other, the network latency and jitter have been significantly reduced. Taihe Interactive's games deployed on another cloud platform used to experience more than 200-microsecond latency. With Virtual Private Cloud, the latency is shortened to below 80 to 90 microseconds worldwide.

The better network quality is reflected in the number of user reports regarding connectivity issues that Taihe Interactive receives. On the other cloud platform, the game developer used to get 30 to 50 user reports per week complaining about network latency and jitter. It required back-and-forth communication with the previous cloud provider to clear a connectivity issue, which normally took one to two weeks. The whole process was very time- and labor-consuming. 10 months after the migration to Google Cloud, only 10 similar cases have been reported. This means that Taihe Interactive's DevOps team can spend less time on troubleshooting.

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Moreover, with a more stable and speedy global network, Taihe Interactive is also able to offer more quality gameplay that was not possible to realize before, such as multiplayer real-time combats.

Thanks to the better connectivity and more appealing features, the user churn rate of Dynasty Legends 2 has dropped by 20%.

"The changes that Virtual Private Cloud has brought us through a speedy, highly connected global network are enormous and multidimensional," says Zhang. "Not only have we significantly reduced the glitches in our online game, but we also have more time and capabilities to develop innovative and exciting gameplay."

## **More reliable and cost-effective VMs supported by Compute Engine**

Zhang says that before the adoption of Compute Engine (<https://cloud.google.com/compute>), the Taihe Interactive team had to manually relaunch one of its VMs one to two times every month, because the machine was too old or encountered errors. It took around 30 minutes each time to relaunch the machine and run through testing, during which the gaming service was disrupted. "The frequency of compulsory VM relaunch was so high that it greatly affected the gaming experience. On Google Cloud, we've only experienced VM relaunch once over the

past 10 months, and it only took 15 minutes for us to make our service available again," he adds. "This not only helps improve the reliability of our services, but also reduces the effort required for VM maintenance."

After Taihe Interactive started employing the VMs on Compute Engine as the game servers of Dynasty Legends 2, and Cloud Storage

(<https://cloud.google.com/storage>) to store data backups and log files, the high performance of the VMs on Compute Engine has helped improve the overall gaming experiences that Taihe Interactive offers. On top of that, as Compute Engine provides custom VMs, Taihe Interactive is able to choose the combination of CPUs and memory for its game servers according to the actual needs. Zhang explains that with fixed VM specifications, the game developer used to pay for unnecessary computing resources, because the computing needs of online games fluctuate hugely. Overall, the team has saved 30% of operational costs with the custom VMs of Compute Engine.

## **Improving security and user interactions**

Another great advantage of migrating to Google Cloud for Taihe Interactive is the improved security. To distribute incoming traffic among its game servers, the game developer leverages Cloud Load Balancing (<https://cloud.google.com/load-balancing>), which also has built-in mechanisms to prevent and mitigate DDoS attacks. Previously, Taihe Interactive once had one of its games down for three hours

right after the release due to a DDoS attack, which was detrimental for a newly launched game. Because of the lack of cost-effective protection mechanisms, Taihe Interactive's games suffered from different levels of DDoS attacks every year. With Cloud Load Balancing, Dynasty Legends 2 hasn't been affected by DDoS attacks since the migration, and the Taihe Interactive team can easily ensure that its services stay uninfluenced when under attack.

Besides security, using Google Cloud also enables Taihe Interactive to improve the interactions among its users and help gamers understand the game content better through Translation AI (<https://cloud.google.com/translate>). As Taihe Interactive's users come from different countries and speak different languages, they could only communicate with basic vocabulary when playing games. With the highly accurate real-time translation across dozens of languages supported by Translation AI, its users can now effortlessly chat for hours and build a stronger bond between each other while engaging in battles together.

"On top of the high-performance infrastructure, Google Cloud also provides tools that allow us to reinforce security and create a more interactive, community-like gaming experience. This has brought the quality of our gaming services to yet another level," says Zhang.

**Unlocking development potential with a more powerful cloud infrastructure**

In 2023, Taihe Interactive will expand the operation range of Dynasty Legends 2 to more countries like Vietnam and South Korea. Moving forward, the team plans to take advantage of the improved global connectivity supported by Google Cloud to develop a new open-world game. It will also adopt more Google Cloud products, such as BigQuery (<https://cloud.google.com/bigquery>), to conduct data analytics for advertising and product development.

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Zhang notes, "We're very impressed by the substantial improvements that Google Cloud has helped us achieve for our game. With the enhanced connectivity and service stability, we're now ready to bring more fascinating gaming experiences to users around the world."